Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Generative AI & LLM Orchestration, Agentic AI Workflows, Prompt Engineering, GitHub Copilot Integration, Autonomous Testing & QA Automation, Predictive Machine Learning, Time-Series Modeling (SARIMA), Data Governance & Lineage, Automated Anomaly Detection, Dimensional Modeling, CI/CD for AI Tools

Programming Languages & Tools: GitHub Copilot, KIRO IDE, Python, PySpark, XGBoost, Scikit-Learn, Azure (ADF, Databricks, Synapse, Azure SQL), AWS (S3, Redshift), Snowflake, Apache Spark, Power BI (DAX), Tableau, Collibra

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Data & AI Systems Engineer - People Analytics & AI Platforms — **May 2022 – Present**

- **Architected and deployed custom autonomous agentic AI workflows** utilizing KIRO IDE, prompt orchestration, and GitHub Copilot to automate schema discovery, data profiling, and code migration, accelerating delivery velocity by 40-50% for enterprise data tools.
- **Engineered automated testing and evaluation frameworks for conversational AI platforms** similar to SpectrumIQ, cutting manual QA turnaround by 50% and establishing rigorous governance, data validation, and automated quality guardrails for production release cycles.
- **Built and tested GitHub Copilot skills and AI tools** that encode complex analytical methods, transforming raw analytical logic into trusted, self-serve data capabilities and directly embedding with internal teams to drive high user adoption.
- **Designed autonomous machine learning anomaly detection frameworks** achieving 95% precision in identifying schema variations, distribution drift, and ETL discrepancies to safeguard data integrity across high-throughput cloud data platforms.
- **Implemented source-to-target reconciliation firewalls and Collibra data lineage** reducing operational turnaround by over 80% while ensuring strict adherence to data governance, confidentiality, privacy standards, and secure enterprise data handling.

Data Scientist & Machine Learning Specialist — **May 2022 – Present**

- **Developed predictive machine learning classification models** using XGBoost, Random Forests, and Scikit-Learn to analyze complex behavioral patterns, delivering actionable telemetry that improved service reliability by 20%.
- **Architected advanced time-series forecasting models (SARIMA)** to predict field support demand and resource allocation, optimizing workforce capacity planning and reducing response times by 20%.
- **Processed massive enterprise datasets at scale** using Apache Spark, PySpark, and distributed SQL pipelines across AWS and Azure environments with 99.9% data availability SLAs.

Business Intelligence & Decision Analytics Specialist — **May 2022 – Present**

- **Created executive Key Operating Metric (KOM) dashboards and semantic models** in Power BI (DAX) and Tableau, translating complex transactional datasets into clear, self-serve strategic insights for corporate leadership.
- **Modeled high-throughput cloud dimensional architectures** utilizing Star/Snowflake schemas and SCD Type 1/2 methodologies within Azure Synapse, Databricks, and Snowflake to streamline enterprise reporting.
- **Partnered directly with cross-functional product groups** to translate operational feedback into actionable product improvements, shaping internal tool roadmaps and scaling analytical capacity across teams.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.